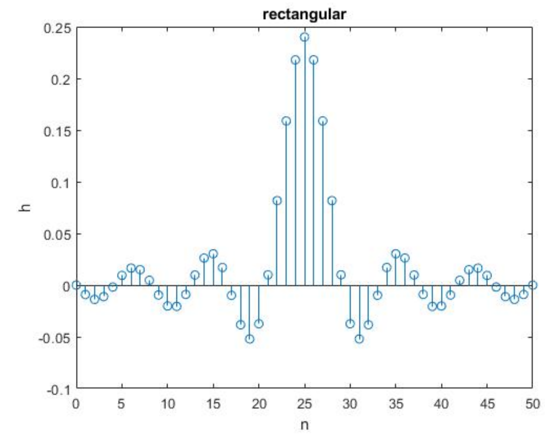
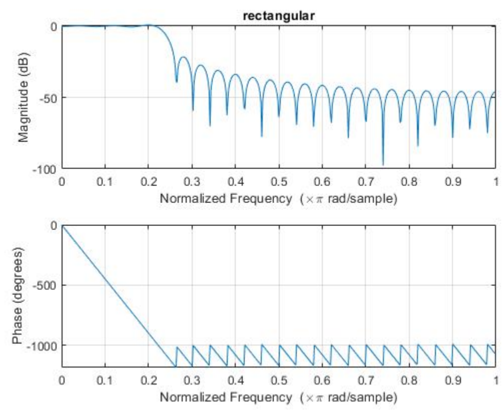
HW6 FIR design by the window method

1.

Define
$h_d[n]$ is a sinc function in time domain named ideal filter.
$w[n]$ is a symmetric window function such as rectangular window, Hann window, Hamming window, and Kaiser window.
$h[n]$ is a finite length filter derived from $h_d[n]w[n]$.

We write a function myFIRrecwin which returns the impulse response $h[n]$ given cutoff frequency omega_c and N. In this question, the cutoff frequency is 0.24π and N is determined by the recommended width of main-lobe (transition width) $4\pi/\Delta\omega/2$ for a rectangular window.
Hence, N is about 25 and the total length of window is 2N+1. Since we have limited the range of h once the n is set, it can be viewed as the ideal filter $h_d[n]$ multiplying a rectangular window function $w[n]$. The resultant impulse response $h[n]$ is shown left below and corresponding magnitude response is shown right below.



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function h = myFIRrecwin(omega_c, N)
% setting range is equivalently implemented
% rectangular window of length 2N+1
n = -N:1:N;
h = sin(omega_c*n)./(n*pi);
h(N+1) = omega_c/pi;
h = h';
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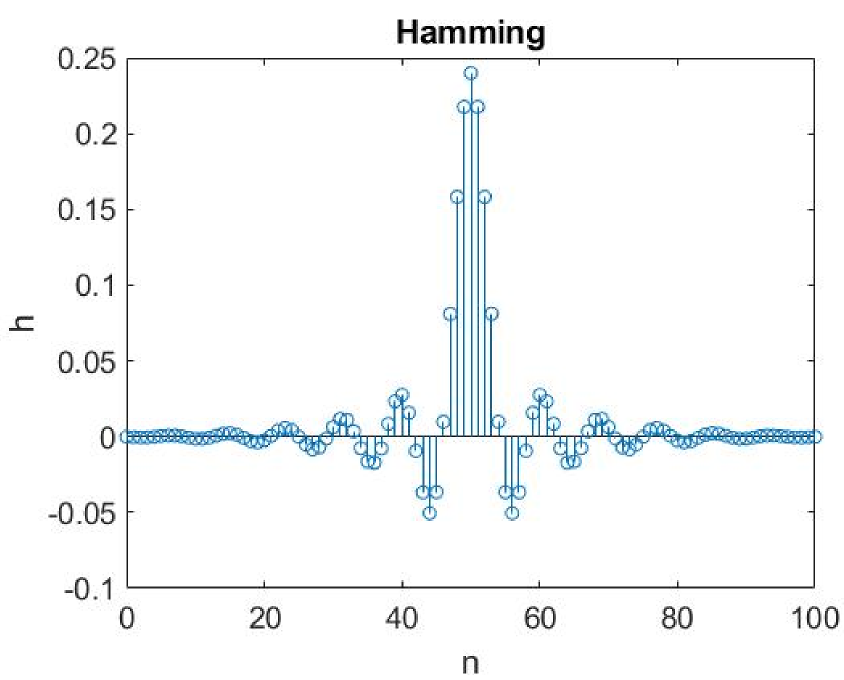
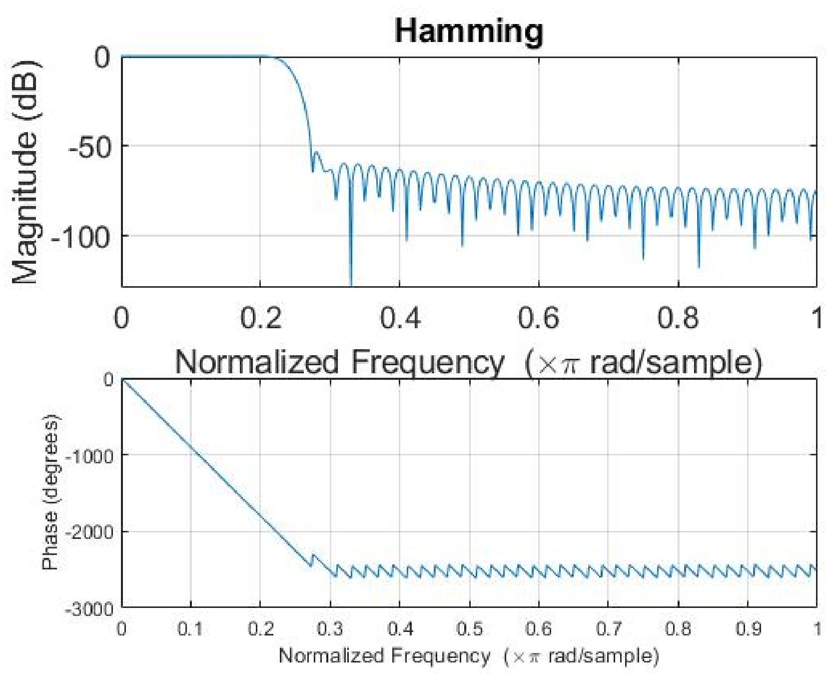
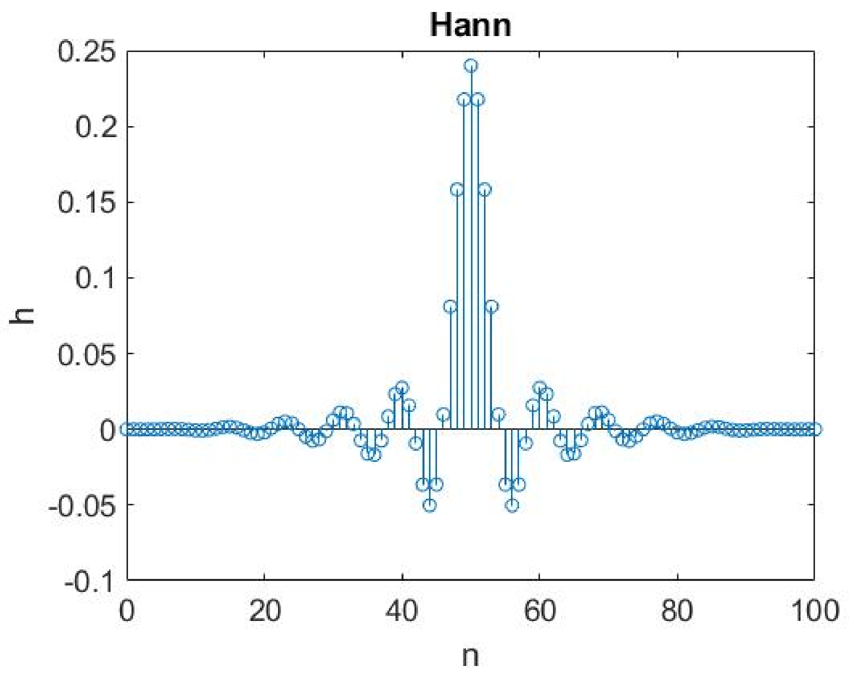
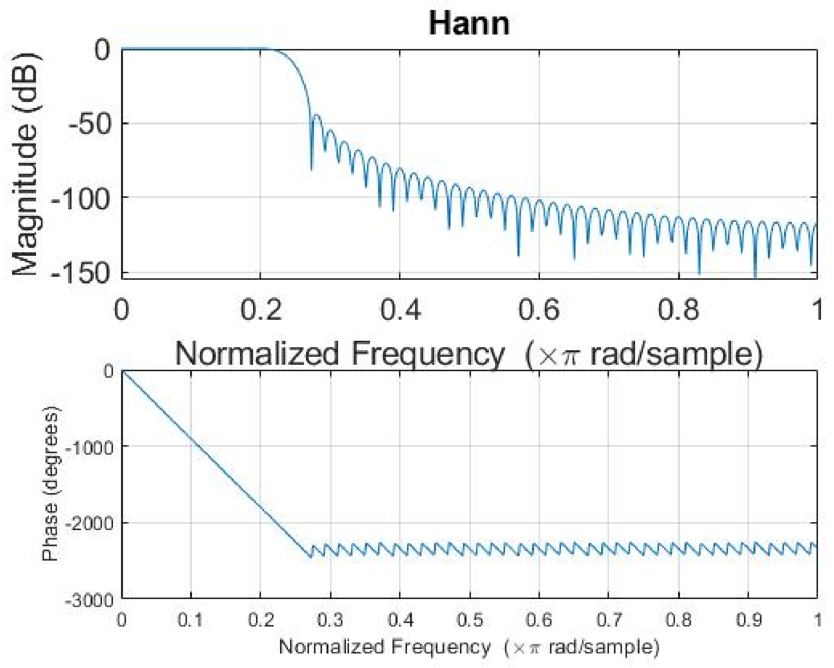
2.

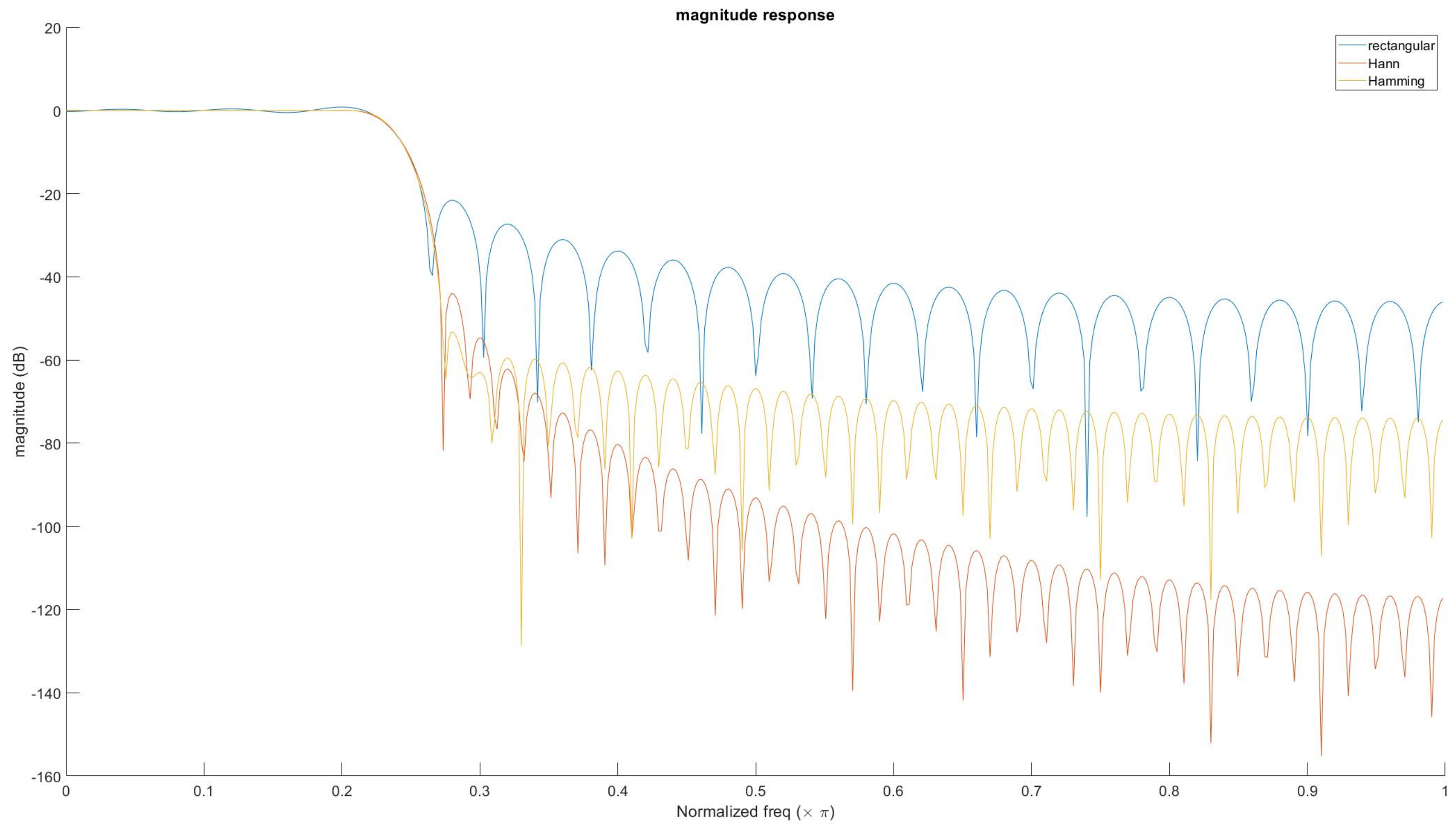
Similarly, we draw the impulse response and magnitude responses of finite length filter windowed by Hann and Hamming window below.











Comparing the three window type, we observe that Hamming window has the most significant drop-off at the first (peak) side lobe. However, at the expense of side-lobe suppression ratio, it has the broadest main-lobe. In contrast, rectangular window has a narrow main-lobe but strong side lobe. Note our desired filter should have narrow main-lobe and weak side lobe as much as possible, but they are always required to make a trade-off. Additionally, some interesting thing seen notably in the trend of side lobe, that of Hann window rolls off persistently differing from others.

3.

Two parameters of Kaiser filter β and α is defined:

$$M = \frac{A-8}{2.285\Delta\omega} => \alpha = \frac{M}{2}, 0 \leq n \leq M$$

$$\beta = \begin{cases} 0.1102(A-8.7), & A > 50 \\ 0.5842(A-21)^{0.4} + 0.07886(A-21), & 21 \leq A \leq 50 \\ 0, & A < 21 \end{cases}$$

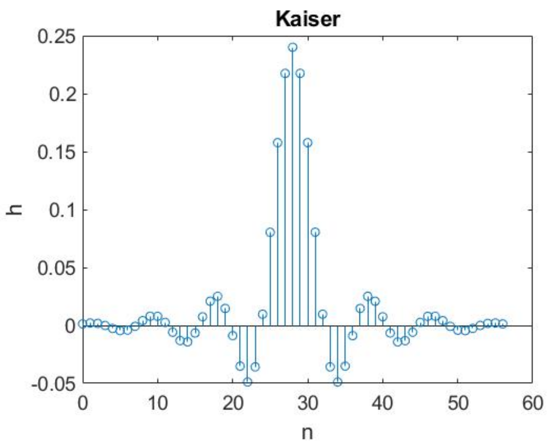
And properties

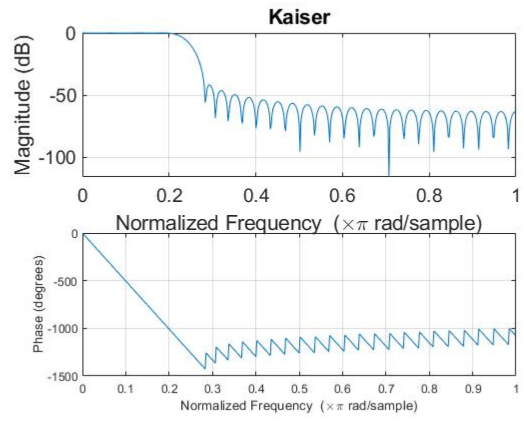
i. If M increases, then main-lobe of magnitude response becomes narrower but side-lobe suppression ratio almost without any change.

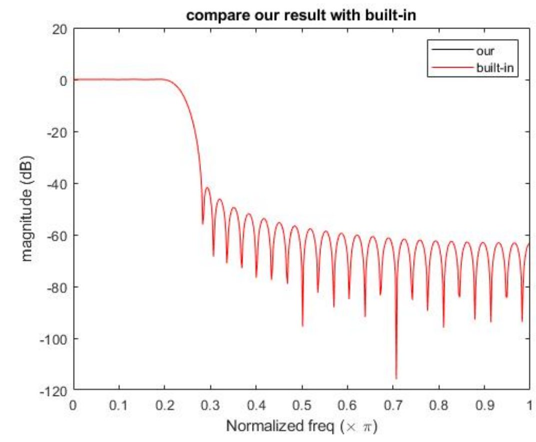
ii. If β increases, then main-lobe of magnitude response becomes broader but side lobes mitigate.

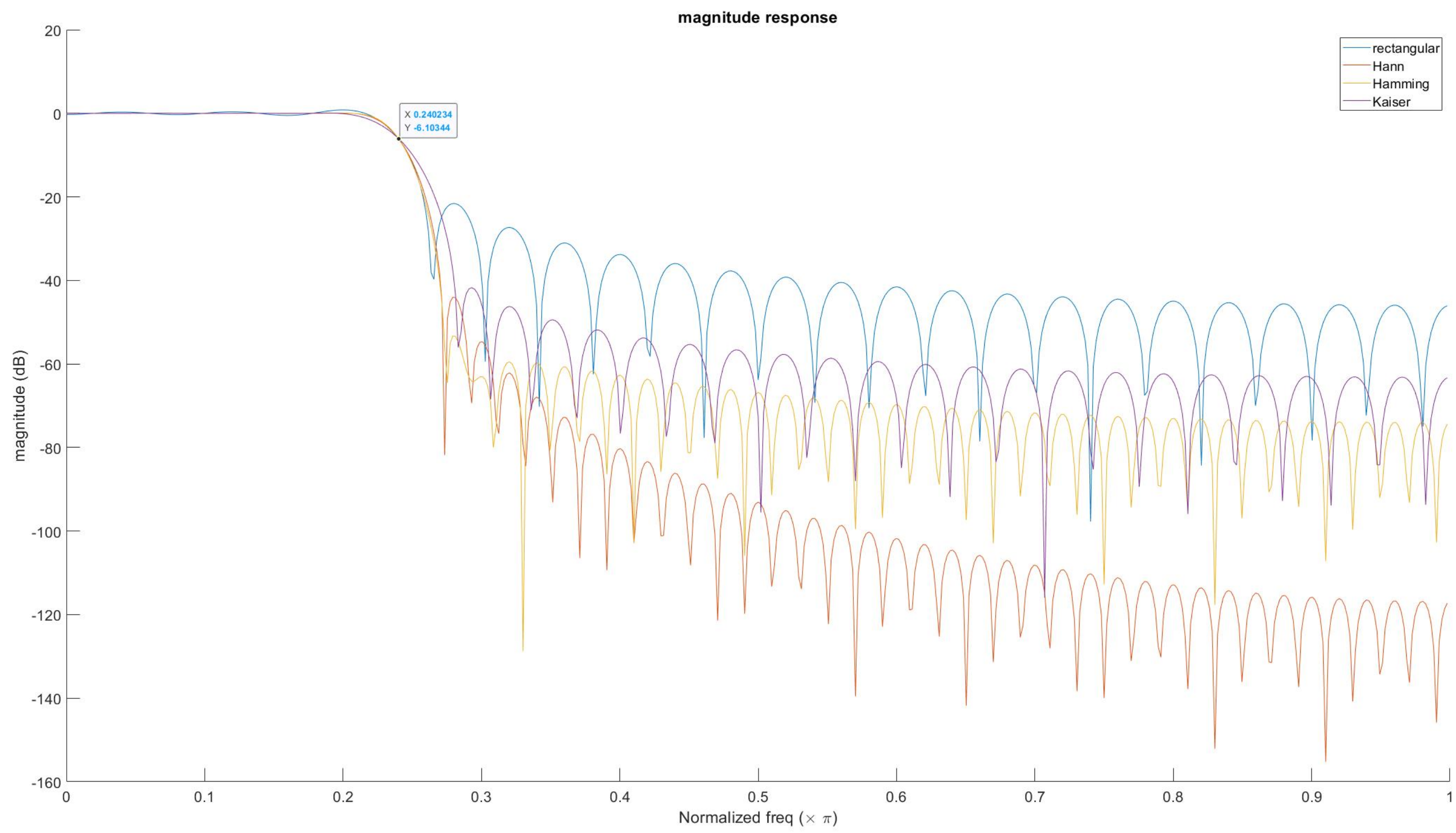
iii. If tolerance δ decreases or A grows, then M and β simultaneously increase.

In this problem, A and $\Delta\omega$ are 40 and 0.08π respectively so M is about 56, namely N = 28, and β is about 3.3953. After applying this Kaiser window, the finite impulse response and magnitude response are shown below. We also write a function called *TestBuildInKaiser* to justify our window.









Four magnitude responses of different windows are drawn in the same figure above. All of them approximately pass through the -6dB point at normalized frequency 0.24π which is exactly the cutoff frequency. The performance of Kaiser window's seems not too extraordinary, because it's an unfair comparison due to the window length. According to Table 7.2 in lecture handout, value A of Hann window and Hamming window are 44 and 53 respectively, while we set value A as 40 here. Let's summarize information to a table at right side and try to interpret why the performance of Kaiser window is worse than Hann and Hamming in this question.

At first, the lack of filter length gives rise to a broader main-lobe due to **property i** as mentioned, so we can see the first side lobe of Kaiser window occurs at the highest frequency among them.

Secondly, recall that β is 3.3953 which is much fewer than the equivalent β value of both Hann and Hamming window. The relatively lower β value leads to stronger side lobe and narrower main-lobe. As the figure indicates the Kaiser window's side lobe is stronger than that of Hann and Hamming window. It's intuitive that the enormous variation of window length dominates the main-lobe width such that it becomes wider comprehensively.

window	filter length (2N+1)	equivalent Kaiser window beta
rectangular	51	0
Hann	101	3.86
Hamming	101	4.86
Kaiser (A=40)	57	-

4.

Once we take into account computational complexity, recall that IIR filter, e.g. Butterworth filter in HW5, needs to do transformation between continuous time and discrete time. After mapping to z domain, we further implement inverse z-transformation and calculate its difference equation. Finally, output signal is processed sequentially from input signal **one by one**. It implies memory is persistently

occupied during process. On the contrary, FIR filter can process signal by efficient algorithm FFT and IFFT that would not occupy memory for a long time. IIR filter suffering from procedure must take much more time on extremely long signal, while FIR filter can use overlapped-and-add method to accelerate process.
Group delay also being a factor might affect our choice. For FIR filter, group delay is a constant. In other words, all phase in different frequencies vary in the same way after passing through this kind of filter. Accordingly, the resultant sound is clear and crispy. However, IIR filter doesn't have consistent group delay, so the variation of phase in different frequencies are also different. It implies that the resultant sound would disperse and timbre would be changed.